Lecture 25

Orthogonal basis $q_1, \dots, q_n$

Orthogonal matrix Q : square, $Q^TQ = I$

Gram-Schmidt $A \longrightarrow Q$

Orthonormal vectors

$$q_i^T q_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$$\underbrace{\begin{bmatrix} - q_1^T - \\ - q_2^T - \\ \vdots \\ - q_n^T - \end{bmatrix}}_{Q^T} \underbrace{\begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & & | \end{bmatrix}}_{Q} = \underbrace{\begin{bmatrix} 1 & 0 & & \\ 0 & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}}_{I}$$

If Q is square, $Q^TQ = I$ tells us that $Q^{-1} = Q^T$.

Examples:

$$\underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\substack{\text{Perm. matrix} \\ Q}} \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{Q^T} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$Q = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

$$Q = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$$

Q has orthonormal columns

- Projection onto its column space:

$$P = Q(Q^TQ)^{-1}Q^T = \underbrace{QQ^T}_{\substack{I \text{ if } Q \text{ is} \\ \text{square}}}$$

$$P^2 = QQ^TQQ^T = QQ^T = P, \quad P^T = P$$

- $A^TA\hat{x} = A^Tb$, $\quad p = A\hat{x}$

Now A is Q $\quad \hat{x} = Q^Tb$

$$\boxed{\hat{x}_i = q_i^T b}$$

$$p = Q\hat{x} = \hat{x}_1 q_1 + \hat{x}_2 q_2 + \cdots$$
$$= q_1(q_1^T b) + q_2(q_2^T b) + \cdots$$

Gram-Schmidt

Indep. vectors

a, b, c

$\downarrow$

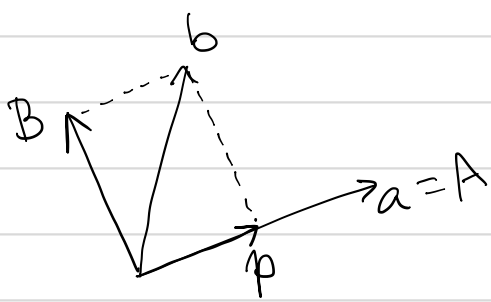
Orthogonal

$A = a, \quad B = b - \frac{A^T b}{A^T A} A, \quad C = c - \frac{A^T c}{A^T A} - \frac{B^T c}{B^T B}$

$\downarrow$

Orthonormal

$q_1 = \frac{A}{\|A\|} \quad q_2 = \frac{B}{\|B\|}, \quad q_3 = \frac{C}{\|C\|}$



$$A^T B = A^T \left(b - \frac{A^T b}{A^T A} A \right) = 0 \checkmark$$

$$a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \qquad A \perp B$$

$$A = a \quad B = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \frac{3}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} | & | \\ q_1 & q_2 \\ | & | \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 1/\sqrt{2} \\ 1/\sqrt{3} & -1/\sqrt{2} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} \qquad \boxed{A = QR}$$

R is upper triangular

$$\begin{bmatrix} | & | \\ a_1 & a_2 \\ | & | \end{bmatrix} = \begin{bmatrix} | & | \\ q_1 & q_2 \\ | & | \end{bmatrix} \begin{bmatrix} q_1^T a_1 & q_1^T a_2 \\ \underbrace{q_2^T a_1}_{0} & q_2^T a_2 \end{bmatrix}$$